

A convergence of space and time: The case of *as long as* and *until* in Czech

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1 Introduction

This chapter provides a discussion of temporal *as long as* and *until*-clauses in Czech, focusing primarily on the syncretism that occurs between the subordinating conjunctions. To illustrate the puzzle at hand, consider the data in (1), as presented in Dočekal (2012).

- (1) a. Petr četl knížku **dokud** Marie zpívala.
Peter read.PST.IMPF book DOKUD Mary sing.PST.IMPF
'Peter was reading a book as long as Mary was singing.'
- b. Petr četl knížku **dokud** se Marie
Peter read.PST.IMPF book DOKUD REFL Mary
neprobudila.
NEG-wake.up.PST.PERF
'Peter was reading a book until Mary woke up.'

As can be seen in (1), both temporal clauses are introduced by *do-kud*, a complex marker composed of the preposition *do* ('up to'), and the relativising wh-element *kud* ('where'), (Dočekal 2012: 190). Given this morphological composition, however, it is slightly surprising that the subordinator can be used in both durative and delimitative contexts. As is exemplified in (2a), the preposition *do* typically only has the delimitative reading in other (nominal) contexts. It also has a directional reading, as in (2b), but crucially, it can never obtain a durative interpretation on its own.

- (2) a. Léto vydrží **do** středy, potom se
summer last DO Wednesday.GEN after REFL
zatáhne a ochladí.
cloud.over and cool.down
'Summer will last until/*during Wednesday, afterwards it
will become cloudy and cold.' [TEMPORAL]

- b. Musí ukrajinské děti chodit v Česku **do**
 have.to.3PL Ukrainian children go.INF in Czechia DO
 školy.
 school-GEN
 ‘Do Ukrainian kids have to go to school in the Czech Republic?’
 [SPATIAL]

The double reading also cannot follow from the denotation of the wh-element, as *kud* merely means ‘where’ in other circumstances as well, (3). These observations raise questions about the relationship between the two instantiations of the subordinator.

- (3) a. **od-kud**
 ‘from-where’
 b. **kud-y**
 ‘Lit: where-through’
 ‘which way’

Based on this apparently exceptional, durative reading of *do-(kud)*, one might want to argue that there are simply two different instances of the subordinator, and that this is, therefore, *not* a case of syncretism, but rather of accidental homophony. However, there are several empirical observations which challenge this view.

Firstly, the morphological identity between the *as long as* and *until* marker in (1) is not unique to Czech. It also occurs in a number of other Slavic languages, suggesting the existence of a stable pattern. Take Slovene, for instance, in (4). To express both types of temporal relations, this language uses the complex subordinator *do-kler*. This marker also comprises the preposition *do* (‘leading up to’), and a wh-element, *kler*, which is derived from the Proto-Slavic interrogative pronoun **kъ* and the emphatic article **lě*, and likely expresses something related to time, e.g. *when*, (Snoj 2016).

- (4) a. Človek se učí, **dokler** živi.
 person REFL learn DOKLER lives.IMPF
 ‘A person is learning as long as he lives.’
 b. Ta prepoved bo ostala v veljavi, **dokler** ne
 this ban will-AUX remain in validity DOKLER NEG
 bodo predloženi novi dokazi.
 will-AUX produce-PASS new evidence
 ‘This ban will remain in force until new evidence is produced.’
 (Dinković & Ilć 2017)

Similarly, in Serbo-Croatian, (5), both clauses are headed by the complex marker *do-k*, which also consists of the preposition *do* ('up to') and a wh-element, *k*, which stems from the same pronoun as Slovene *kler*.

- (5) a. Solarno punjenje **dok** je uređaj isključen.
solar charging DOK is device switched-off
'Solar charging as long as the device is shut off.' (Internet)
- b. Skrivat ću se **dok** (ne) svane dan.
hide-1sg fut refl DOK (NEG) dawn-3sg.pres day.
'I will hide until it dawns' (Dinković & Ilć 2017)

However, even though the Slovene and Serbo-Croatian examples indicate that we are not dealing with an isolated case, they do not conclusively rule out that the subordinators are homophonous, because they are all cognates.

A second piece of evidence in favour of syncretism, however, is the fact that the morphological identity persists in several other Slavic languages, independently of any specific morphological realisations (i.e., the subordination is marked by a different exponent than the preposition *do*). Russian, for example, (6), features the complex subordinator *po-ka*, which is composed of the (predominantly locative) preposition *po* ('along, after, up to'), and the Old Church Slavonic determiner and relative pronoun *ki* ('which'), (Eckhoff & Berdicevskis n.d.).

- (6) a. **Poka** ya zhil v Rossii ya dumal v Anglii odni
POKA I lived in Russia I thought in England only
migranty zhivut
migrants live
'As long as I lived in Russia, I thought that only migrants lived in England.' (Internet)
- b. Ya podozhdu **poka** on ne vernotsya.
I wait.PERF.FUT POKA he NEG come.back.PERF.FUT
'I will wait until he comes back.' (Wade 2011)

In Polish, (7), we also find the complex marker *po-ki*, albeit mainly in the literary register. This subordinator bears strong resemblance to the Russian one in that it is also made up of the preposition *po* ('on, after, up to'), and a wh-element, *ki*, which also stems from the same determiner/pronoun as Russian *ka*, (Brückner 1985).

- (7) a. Wejdź na pokład **póki** możesz.
come.IMPER on board PÓKI can.2SG
'Get on board as long as you can.' (Internet)

- b. Żyli długo i szczęśliwie, póki nie
 lived-PST.3PL long and happy PÓKI NEG
 umarli.
 died.PST.PF.3PL
 ‘They lived happily ever after until they died.’ (Internet)

Slovak, (8), is even more interesting in this regard. Even though multiple strategies are available to express *as long as* and *until* in this language - one of which also involves the preposition *po* (cf. Section 3) - one of the most commonly used ones today is the grammaticalised marker *kým*, which formally corresponds to the instrumental form of the interrogative pronoun *kto* (‘who’). Here too, the syncretism is preserved, despite the fact that the marker is morphologically unrelated to the delimitative/directional one in Czech.

- (8) a. **Kým** sme mladí, všetko sa nám zdá krajšie
 KÝM we-are young everything to us seems prettier
 ‘As long as we are young, everything seems more beautiful to us.’
 b. Čakal, **kým** neprestane snežiť.
 wait.PST.IMPF KÝM NEG-stop.PST.PERF snow
 He was waiting until it stopped snowing.
 (Slovník Jazykovedného ústavu Ľ. Štúra SAV)

Regardless of the exponents that languages seem to employ, *as long as* and *until* seem to cluster together consistently. In fact, a final argument against an explanation based on accidental homophony is the observation that this clustering is not specific to Slavic languages either, but constitutes a rather widespread phenomenon cross-linguistically (cf. Wälchli (2018) for a more comprehensive overview). To give an example from a language that does not belong to the same language family, consider the Hungarian data in (9). The sentences feature the complex subordinator (*amedd*)-*ig*, which is formed with the demonstrative pronoun *a(z)*- (‘that’), the wh-element *medd*- (‘how far’), and the terminative case suffix *-ig* (‘until’).

- (9) a. Itthon maradok, **ameddig** Emma munkában van.
 home I-stay AMEDDIG Emma work-in is
 ‘I will stay home as long as Emma is at work.’
 b. Itthon maradok, **ameddig** Emma nem jön haza.
 home I-stay AMEDDIG Emma NEG comes home
 ‘I will stay home until Emma comes home.’ (Ürögdi 2013)

The pervasiveness of this phenomenon leads me to conclude that there is an underlying pattern to be investigated. The central question this chapter will revolve around is thus how two contrastive expressions, i.e., durative *as long as* and delimitative *until*, relate to each other - especially because this relation is suggested by the cross-linguistically recurring syncretism between the subordinators, even though it cannot be immediately deduced from the (morphological) sum of their parts.

A few of the previous accounts indirectly provide an answer to this question, (cf. Section 2), each of them arguing for a certain default reading, and connecting the ambiguity either to other clause-internal elements, such as negation or verbal aspect, or to different scope positions of the subordinator itself. I will briefly review these accounts to provide some background into the issue. I will also point out several predictions they make that are not borne out with respect to Czech.

Drawing on the alternative *po*-pattern in Russian, Polish and Slovak, (cf. Section 3), I will therefore diverge slightly from the literature, and propose that the key to understanding the relationship between *as long as* and *until* could also lie in the morphological connection these markers still have with the nominal domain. More concretely, I will draw a parallel between the spatial and temporal domain and show that the ambiguity can also be traced back to the well-known alternation between locative and directional readings of simple adpositions in the spatial domain, (e.g., He was sitting **on** the chair vs. He sat down **on** the chair). After establishing this parallel, I will show that the syncretism can be captured by assuming a typical Nanosyntactic containment relationship (cf. Caha et al. in press, Starke 2009). For this, I will build on Pantcheva's (2011) work on directional expressions.

After laying out the general idea of the analysis, I will return to Czech, and discuss how the proposed solution with the *po*-marker can be extended towards *do*. Although there is no full syncretism in Czech, covering both the spatial and temporal domain, the pattern can still be explained by means of the same model of lexicalisation.

Finally, (cf. Section 5), I will summarise the chapter and briefly provide an avenue for future research.

2 Previous accounts

There has been a long and lively debate in the literature about how many *untils* there are in the lexicon (cf. Mittwoch 1977, Karttunen 1974, De Swart 1996, Giannakidou 2002, Condoravdi 2008 amongst others). This debate is generally based on the well-known contrast in English between pairs of sentences like *John slept/didn't sleep until 5pm* and **John arrived until 5pm/John didn't arrive until 5pm*. However, the issue of why the same marker is also frequently used across languages to introduce *as long as*-clauses has not received as much attention. To

the best of my knowledge, there are only two types of formal accounts which provide an explanation for it in their analysis. In one of those accounts, the different readings are attributed to the semantic interaction of the subordinator with negation and the aspect of the embedded predicate. In the other one, it is argued that the readings arise through different scope positions of the subordinator in syntax. In this section, I will briefly go over each of these accounts before proceeding to a discussion of the *po*-pattern in Russian, Polish and Slovak, which will serve as support for an alternative approach.

2.1 Interaction with negation and aspect

For Czech specifically, the current issue has been addressed by Dočekal (2012), mainly from a semantic perspective.¹ He argues that there is only one type of temporal clauses, namely the durative *as long as* type, and that the other interpretation is a mere by-product of the interaction between *dokud*, negation and aspect.

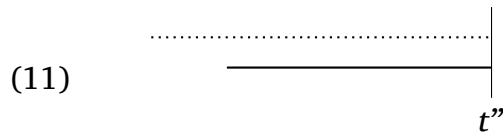
Concretely, he proposes that the lexical entry of *dokud* is durative in nature. It is essentially a binary operator which takes the matrix clause P and the embedded clause Q as its arguments, and imposes, on the one hand, that the running time of P be a proper subset of the running time of Q, and on the other, that both events end at a dummy time *t*”, which corresponds to the right boundary of both events.

To better understand this, I visualise the relation between P and Q in (11) for the sentence in (1), repeated here for convenience in (10).² The solid line represents the matrix event P and the dotted one the embedded event Q. As can be seen, the figure demonstrates the requirement encoded in the lexical entry of *dokud*: the event of Peter reading the book must take place during the time interval that Mary is singing, and when one of them stops their activity, the other must do as well.

- (10) Petr četl knížku, dokud Marie zpívala.
 Peter read.PST.IMPF book as-long-as Mary sing.PST.IMPF
 ‘Peter was reading a book as long as Mary was singing.’

¹Dočekal’s proposal is reminiscent of Tovená’s earlier treatment of Italian *finché* ‘as long as, until’. For reasons of space, I will not provide a full discussion of her account as well, but it is worth pointing out that the main difference between their analyses lies in the amount of influence the subordinator exerts on the interpretation of the temporal clause. For Dočekal, the subordinator functions as an operator which actively puts restrictions on the clauses it selects, and which, consequently, acts as the driving force behind the two readings. For Tovená, however, it seems that the primary cause of those readings is negation (and to some extent also aspect), as it affects the temporal boundaries which the subordinator ends up selecting in each case. The subordinator thus seems to have a more passive role in her analysis.

²The visualisations are based on those in Tovená’s (1996) dissertation, but I adjusted them slightly to suit the data in the chapter.



In order for P to always form a subset of Q, *dokud* also requires that the predicate of the embedded clause is ‘homogeneous’. This means that it must have a subinterval property, or, put differently, that it has to be atelic or durative.

If this requirement is not immediately satisfied - for example, in the case of perfective predicates - Dočekal expects negation to intervene to turn the predicate into a compatible argument of *dokud*. This is illustrated in (12).

- (12)
- a. Petr zpíval, dokud Marie hrála.
Peter sing.PST.IMPF as-long-as Mary play.PST.IMPF
‘Peter was singing as long as Mary was playing.’
 - b. *Petr zpíval, dokud Marie
Peter sing.PST.IMPF as-long-as Mary
dohrála.
finish.playing.PST.IMPF
‘Peter was singing as long as Mary finished playing.’
 - c. Petr zpíval, dokud Marie
Peter sing.PST.IMPF as-long-as Mary
nedohrála.
finish.playing.PST.IMPF
‘Peter was singing as long as Mary had not finished playing.’

While in (12a), both events span over a large enough time interval that P can be contained within Q, this is not the case in (12b). There, Q is a punctual event which corresponds to either a single point in time or a very short time interval. As such, it does not naturally possess the subinterval property and does not match with the subordinator. This incompatibility is remediated by the presence of the negative marker *ne-* in (12c).³ The sentence is then reinterpreted as follows: the event of Peter singing holds during the time interval that Mary has *not* finished.

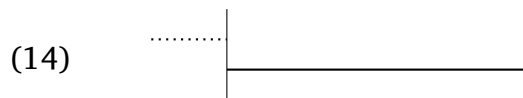
Note that cases such as (12c) thus do not qualify as real *until*-clauses for Dočekal. They rather behave as duratives, as in (10), i.e., both clauses end at the same point in time, and both clauses are understood

³In Mittwoch (1977) and subsequent work, it has been argued that negation acts as a stativising operator, turning the event into a negative state. Dočekal points out that negation basically provides the subinterval property for free through downward entailment, and that there is, therefore, no need to postulate any additional functions. Regardless of the approach, though, the result is fundamentally the same for both: negation affects the properties of the time interval of the embedded event.

to be coterminous. In fact, for him, any *until*-reading is a derivative side-effect of the main, durative reading, which, in addition, only surfaces when the sentence has a negated matrix clause, creating the so-called ‘actualisation inference’. This inference refers to the fact that the matrix event is implicated to take place at the point in time when the embedded clause reaches its culmination point (e.g., recall English *John didn’t arrive until 5pm* → when the clock strikes 5, John must arrive for the sentence to be felicitous.). In other words, Dočekal considers only those clauses where the events can be ordered consecutively to be proper *until*-clauses.

This claim can be illustrated by means of the example in (13). It is also visualised by the figure in (14). The negator in the matrix clause is responsible for the additional interpretation that the event of the princess marrying Honza cannot hold true *before* the event of him killing the dragon, or more abstractly, that the realisation point of Q (dotted line) forms the starting point of P (solid line).

- (13) Princezna si Honzu nevzala, dokud nezabil draka.
 princess REFL Honza NEG-married until NEG-killed dragon
 ‘The princess didn’t marry Honza until he killed the dragon.’



Now, while Dočekal’s account seems to be able to capture all the Czech facts in a uniform manner, i.e., there is a single lexical item for *dokud* which has a default durative reading and which expresses other readings by engaging with (mainly) negation and verbal aspect, there are also a few inconsistencies to his story.

The first one pertains to his treatment of sentences like the one in (12c). Due to the presence of negation, he considers them to be negative equivalents of positive *as long as*-clauses (i.e., *as long as not*). Under this view, though, we might expect that (i) we would not encounter actually negative and purely atelic *as long as*-clauses, and that (ii), if we *do* encounter them, they would behave in a similar way as the homogeneous, perfective ones. However, neither of these predictions are borne out.

As shown in (15), the example where the embedded, atelic predicate is negated is both acceptable and interpretable, and on top of that, there is a clear distinction in the Negative Concord Item (NCI)/Negative Polarity Item (NPI)-licensing properties of *ne*-.

- (15) a. Petr zpíval, dokud **nikdo**
 Peter sing.PAST.IMPF as-long-as nobody
neprotestoval.
 NEG-protest.PAST.IMPF
 ‘Peter was singing as long as nobody protested.’
- b. *Petr zpíval, dokud **nikdo**
 Peter sing.PAST.IMPF as-long-as nobody
nepřišel.
 NEG-came.PAST.IMPF
 ‘Peter was singing as long as nobody came.’
- c. Petr zpíval, dokud **někdo**
 Peter sing.PAST.IMPF until somebody
nepřišel.
 NEG-finish.playing.PAST.IMPF
 ‘Peter was singing until somebody came.’

Whereas an NCI/NPI-item like *nikdo* ‘nobody, anybody’ is allowed in (15a), the same item (and by extension any other NCI/NPI) cannot appear in clauses where the predicate is negated and perfective without yielding an ungrammatical sentence, as illustrated by (15b). In such cases, it must be replaced by the positive alternant, *někdo* ‘somebody’, as shown in (15c). In other words, *ne-* loses its ability to license NCI/NPIs in this particular context. Despite the fact that Dočekal explains this change in behaviour by assuming a covert movement of the negator to a high C-position (in accordance with Abels (2005)), this discrepancy suggests that perfective negated clauses are not perfect equivalents of negative *as long as*-clauses - that *until* is still a different creature from *as long as*.

A second, but related inconsistency concerns the analysis of the clauses that Dočekal *does* conceive of as *until*-clauses, namely the consecutive ones as in (13). In his view, these sentences behave in a similar fashion as English *John didn’t arrive until Jane left*, but this equation is inaccurate due to the obligatory presence of the negative marker in the Czech embedded clause. As prescribed by his own lexical requirements of *dokud*, negation of the perfective, embedded predicate must give rise to the subinterval property, which means that the resulting reading in his analysis can be never be one where the endpoint of Q forms the starting point of P. It is always going to be a coterminous one instead, as shown in (16). The event of the princess not marrying Honza coincides with that of the *not* killing of the dragon, which means we are back at the scheme in (11).

- (16) Princezna si Honzu nevzala, dokud nezabil draka.
 princess REFL Honza NEG-married until NEG-killed dragon
 ‘The princess didn’t marry Honza as long as he didn’t kill the dragon.’

And yet, the intuition that cases like (16) should also be *until*-clauses seems correct, given the fact that they exhibit the same licensing properties as in (15). If we wish to replace *draka* ‘dragon’ in (16) with the NCI/NPI *nikdo* ‘nobody, anybody’, the resulting sentence is also ungrammatical, as shown in (17). This again highlights the difference between the two temporal clauses, and also casts doubt over the major role that negation plays in all of this, because it not only seems to be semantically non-negative, but relying on it too much leads to the wrong outcomes in the cases presented here.

- (17) a. *Princezna si Honzu nevzala, dokud **nikoho**
 princess REFL Honza NEG-married until nobody
 nezabil.
 NEG-killed
 ‘The princess didn’t marry Honza as long as he didn’t kill nobody.’
 b. Princezna si Honzu nevzala, dokud **něko**
 princess REFL Honza NEG-married until somebody
 nezabil.
 NEG-killed
 ‘The princess didn’t marry Honza as long as he didn’t kill somebody.’

A third and final remark is that Dočekal takes *as long as* to be the default interpretation of the subordinator, explaining away potential *until*-readings by looking at the direct environment in which it occurs. This is, to some extent, unexpected when taking into account the morphological properties of the subordinator that I pointed out in the Introduction, i.e., that *do* yields a delimitative reading in the nominal domain, irrespective of the context in which it appears. Considering that *do* retains those properties even in other complex relativising constructions, such as *dokdy*, (18), one could argue that, if there is supposed to be a default reading, it is *until* rather than *as long as*.

- (18) Měli byste vědět, **dokdy** musíte vyhotovit
 must-PAST.PL AUX-2PL know until-when must-2PL prepare
 účetní závěrku.
 financial statements.
 ‘You must know until when you must have your financial statements prepared.’ (Internet)

In the next subsection, I will discuss Ürögdi's (2013) proposal for Hungarian, which assumes exactly this - that *until* is the base reading from which *as long as* can be derived through movement from a different syntactic position in the clause.

2.2 Scope

As was briefly mentioned in the Introduction, Hungarian is one of the non-Slavic languages in which the observed syncretism between the *as long as* and the *until*-marker can be found. However, the picture is slightly more complex than how it was presented in (9). There are, in fact, varieties of Hungarian which distinguish not just two, but *four* syntactically distinct constructions with (*amedd*)-*ig* as the subordinating conjunction. An example of each is shown in (19) - the examples in (19a) and (19c) are repeated from the Introduction.

- (19) a. Itthon maradok, **ameddig** Emma munkában van.
 home I-stay AMEDDIG Emma work-in is
 'I will stay home as long as Emma is at work.'
- b. Itthon maradok, **ameddig** Emma haza jön.
 home I-stay AMEDDIG Emma home comes
 'I will stay home until Emma comes home.'
- c. Itthon maradok, **ameddig** Emma nem jön haza.
 home I-stay AMEDDIG Emma NEG comes home
 'I will stay home until Emma comes home.'
- d. Itthon maradok, **ameddig** Emma haza nem jön.
 home I-stay AMEDDIG Emma home NEG comes
 'I will stay home until Emma comes home.' (Ürögdi 2013)

Even though the formal differences between the sentences in (19) are minimal - especially between the ones where the subordinator receives an *until*-reading - Ürögdi (2013) remarks that their interpretations are not entirely identical. Only (19b) and (19d) are understood as true *until*-constructions, (19a) and (19c) are both durative. In her study, she thus investigates where the semantic nuances come from and how to account for them, but, unlike Dočekal, she does this without relying on negation or verbal aspect. Instead, she considers the (un)availability of what she refers to as 'high' and 'low' readings, i.e., two readings which are associated with, and dependent on, the scope position of the subordinator itself. She also assumes that if there is a base reading for (*amedd*)-*ig*, it must be *until* - presumably, because the suffix -*ig* unambiguously expresses terminative case, just like Czech *do* is also exclusively used as a directional (cf. (2)).

Concretely, Ürögdi starts from the observation that only the high reading is available for the sentences in (19b) and (19d). This is not immediately obvious, but becomes more apparent when the sentences are turned into multiple embedding constructions, as in (20) and (21). By adding an extra event, it is revealed that (*amedd*)-*ig* can scope over the highest one, i.e., *say*, but has no access to the lower one, i.e., *Emma comes home*.⁴

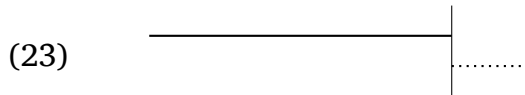
- (20) Itthon maradok, ameddig mondod, hogy Emma haza jön.
home I-stay until say-PRES that Emma home come
HR: ‘I will stay home until the time that you are saying that Emma will come home.’
*LR: ‘You tell me that Emma will arrive by time *t*. And I will stay home until *t*.’
- (21) Itthon maradok, ameddig mondod, hogy EMMA haza nem jön haza.
home I-stay until say-PRES that Emma home NEG jön haza.
come
HR: ‘I will stay home until the time that you are saying that it is not EMMA who will come home.’
*LR: ‘You tell me that it is not EMMA who will arrive by time *t*. And I will stay home until *t*.’

Since the low reading is out for these sentences, Ürögdi argues that they behave like so-called ‘event-relatives’, i.e., constructions where the embedded clause functions as the relativised argument of the matrix event (cf. Lipták 2005, Haegeman & Ürögdi 2010a,b, Kiss 2024 amongst others). As such, the subordinator is considered to be base-generated in a high position just outside of the embedded clause. From there, it functions in a way that feels intuitive for *until*-clauses: it modifies the duration of the matrix event by setting the embedded one as its final point. For the examples in (20) and (21), this means that my staying at home lasts exactly until the point in time when Emma comes home - her arrival cuts off the matrix event. Ürögdi represents the underlying structure of these sentences as follows, (22): *until* originates in a position outside of the embedded clause, and selects a time point *t_i* which corresponds to the (short) time interval of the embedded event.

⁴With respect to the construction in (19d) where negation surfaces in an unusual position between the particle and the verb, Ürögdi shows that there are focus effects at play. For this reason, *Emma* is put in small caps in (21) - the constituent receives emphasis, and the event is interpreted as EMMA not coming home, not someone else. However, this is not a critical of the core of her analysis, so I will not elaborate on this further.

- (22) [I will stay home until t_i [t_i [Emma comes home / does not arrive]]]

This can also be depicted more abstractly, reusing the arrow-like representations from the previous subsection. The event-relative then corresponds to the consecutive type of *until*-clauses we have seen before in (14), but the ordering is reversed, as shown in (23). It is now the matrix event P (solid line) which leads up to the embedded event Q (dotted line), and at the point where they intersect, P ends.



Conversely, for the remaining sentences in (19a) and (19c), Ürögdi notes that, in addition to the high reading, the low reading is also available. She again demonstrates this by turning the sentences into multiple embedding constructions, as in (24) and (25). This time, (*amedd*)-*ig* is able to scope over the highest event, i.e., *say*, as well as the lowest event, i.e., *Emma is at work* / *Emma does not come home*.

- (24) Itthon maradok, ameddig mondtad, hogy Emma munkában
home I-stay until say-PRES that Emma work-in
van.
is
HR: ‘I will stay home as long as you are saying that Emma is at work.’
LR: ‘You tell me that Emma will be home as long as time t . And I will stay home as long as t .’
- (25) Itthon maradok, ameddig mondtad, hogy Emma nem jön
home I-stay until say-PRES that Emma NEG come
haza.
home
HR: ‘I will stay as long as you are saying that Emma will not come home.’
LR: ‘You tell me that Emma will not come home as long as time t . And I will stay as long as time t .’

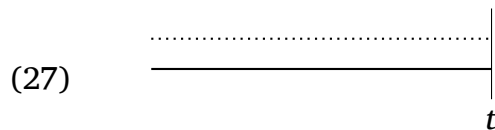
Due to the availability of the low reading, Ürögdi rather refers to these clauses as temporal relatives, i.e., constructions like *the time when*, where what is relativised is the temporal interval relevant to the embedded clause. In these constructions, *until* is base-generated in a low-position inside the embedded clause. From there, it first modifies the duration of the embedded event by determining its endpoint t . After that, it moves

upstairs (in a similar fashion as *wh*-elements) to the higher position outside of the embedded clause. Once displaced, it modifies the duration of the matrix event as well and aligns its final point to the same time point *t*. Because of this double modification, the subordinator adopts the durative meaning that is typical of *as long as*-clauses. Both events last until *t* and are therefore in full or at least partial overlap. For the examples in (24) and (25), this means that the event of Emma being at work or not coming home coincides with my staying at home, and both events end simultaneously.

The underlying structure Ürögdi assigns to these durative constructions thus looks slightly different, as shown in (26). Instead of selecting a single time point t_i that corresponds to the embedded event as a whole, *until* now only takes its endpoint and matches it with the endpoint of the matrix event. It does so by moving out of the embedded clause to the higher scope position.

- (26) [I will stay home [until t_i] [Emma is at work / does not come home [until t_i]]

Visually, this can be represented by the same figure as in (11), repeated here in (27). Without making any particular claims about whether one event ought to be a subset of the other, the matrix event P (solid line) ends at time point *t*, and the embedded event Q (dotted line) ends simultaneously, making them coterminous.



As the structures in (22) and (26) indicate, Ürögdi's proposal successfully differentiates between the various types of (*amedd*)-*ig*-clauses by means of the syntactic position the subordinator is capable of occupying within the clause alone. The other clause-internal elements do not play a big role in her account. In fact, she provides several arguments for why their impact on the temporal interpretation is less significant.

She argues, for instance, that whenever the negator *nem* shows up, it merely expresses sentential or predicate negation and does not affect any properties of the embedded event (e.g., stativising it or contributing a subinterval property). The reason for this, as she explains, is that this would require the stipulation of an exceptional use of negation that does not occur elsewhere. To back this up, she discusses, amongst others, the diagnostic in (28).

- (28) a. # John drops the book.

- b. # John doesn't drop the book.
- c. John owns a car/ John is dancing.

While the predicates in (28a) and (28b) cannot be uttered out of the blue, the static and durative ones in (c) are felicitous statements. If negation altered the properties of eventive predicates in such a way that they became statives or duratives, then (28b) would be expected to pattern with (28c), contrary to fact. This suggests that negation did not transform the event *drop the book* into a real stative or durative.

Moreover, she points out cases where the perfective aspect of the verb does not affect the compatibility with *until* (both in English and Hungarian). Consider the contrast in (29).

- (29) a. *John arrived until 3pm.
 b. John missed the note until 3pm.

Provided that the perfective event can be interpreted iteratively, *until* can combine with it without the interference of negation. While it makes no sense for John to arrive repeatedly, it is perfectly imaginable that he is playing a music instrument and keeps missing a particular note until a given moment in time. Consequently, (29a) must be negated to be grammatical, whereas (29b) is fine without. Similarly, perfective events can appear in embedded clauses as well, as indicated by (19b). Ürögdi thus concludes that there is no reason to assume that negation or verbal aspect are determining factors.

Now, while this is yet another promising account of the ambiguity between *as long as* and *until*, i.e., there is a single delimitative subordinator and the two readings are distinguished by means of its position in the underlying structure of each temporal clause, some of the specifics of both Hungarian and Czech make it hard to map the analysis from one language onto the other. In particular, the biggest issue arises with the derivation of the durative constructions.

As we have seen, Ürögdi argues that the *as long as*-reading is obtained through the movement of the subordinator from a low embedded position to a high matrix position. Support for this movement comes from the fact that, for certain Hungarian variants, (*amedd*)-*ig* can be interpreted in both of those positions, something which is made evident by adding more embeddings to the sentence.

In order to adopt this analysis for the Czech data as well, it must thus be shown that *dokud*-clauses carrying a durative interpretation also allow for the high and low reading. Judging by speaker intuitions, however, this does not seem to be the case. When we apply the same multiple embedding construction test on a purely atelic example, it turns out that the low reading is unavailable. Consider the paraphrases in (30).

- (30) Petr pracoval, až dokud jsi říkal, že
 Peter worked.IMPf up.to as-long-as AUX-you said.IMPf that
 Marie zpívala.
 Marie sang.IMPf
 HR: ‘Peter was working as long as you were saying that Mary
 was singing.’
 *LR: ‘You said that Marie was working as long as *t*. Peter was
 working as long as *t*.’

If the availability of these readings indeed reveals the syntactic positions of the subordinator, and consequently, the type of clause we are dealing with, then the paraphrases in (30) seem to suggest that atelic *dokud*-clauses correspond to Ürögdi’s event-relatives instead, the constructions which allow just one reading and act like genuine *until*-clauses. This is, of course, a wrong outcome, as atelic temporal clauses are also unmistakably durative in Czech. In other words, whereas with Dočekal’s proposal, it proved to be difficult to derive Czech *until*-clauses due to the function assigned to negation, we are now facing the opposite problem with Ürögdi’s proposal; we cannot derive Czech *as long as*-clauses, as the required low position for this reading is not accessible.

To sum up, I have summarised the two accounts which (inadvertently) tackle the issue of the syncretism between *until* and *as long as* in this section. Despite their contrastive approaches, they both fundamentally argue for an explanation based on identity. Dočekal (2012) builds his proposal on the premise that *dokud* is durative, and clarifies how this base property can be expanded towards the delimitative reading. Ürögdi (2013), on the other hand, opts for the opposite premise, and conceives of (*amedd*)-*ig* as a default delimitative marker which acquires the durative reading in the process of syntactic movement. However, while both accounts offer a clear solution for one of the interpretations, they also run into issues to derive the other one — particularly when considering the Czech data.

At this point, I should stress that the aim of this chapter is not to outright challenge these accounts or to address their shortcomings. but rather to offer an alternative perspective on the issue and, in doing so, hopefully shed a different light on it. To this end, I will first make a small detour in the next section to the *po*-pattern that appear in Russian, Polish and Slovak, and I will build on the function of the preposition to connect the clausal marker to the spatial domain.

3 Insights from the *po*-pattern

In this section, I will take a different approach to the data by examining them from a morphological perspective. Instead of assuming that

dokud has a single reading from which the other must be derived - either through various interactions with other clause-internal elements or by movement of the subordinator itself - I will explore the idea that the ambiguity arises through syncretism, i.e., the phenomenon whereby two distinct morphosyntactic features are realised by a single lexical item. More specifically, I will draw on the nominal distribution of the *po*-marker in Russian, Polish and Slovak to argue that those two features correspond to a locative (Place) and directional (Path) head (cf. Section 3.1), and using this as a basis, I will establish a parallel between the spatial and temporal domain (cf. Section 3.2).

3.1 *Po* as a spatial marker

As was hinted at in the Introduction, Russian, Polish and Slovak differ from Czech in that the subordinating conjunction they use to introduce *as long as* and *until*-clauses is formed with the preposition *po* ‘along, after, on’ rather than the preposition *do*. The relevant examples are repeated in (31)-(32), with the addition of Slovak in (33).

- (31) a. **Poka** ya zhil v Rossii ya dumal v Anglii odni
 as-long-as I lived in Russia I thought in England only
 migranty zhivut
 migrants live
 ‘While I lived in Russia, I thought that only migrants lived in England.’
 b. Ya podozhdu **poka** on ne vernotsya.
 I wait.PERF.FUT until he NEG come.back.PERF.FUT
 ‘I will wait until he comes back.’

(Russian: Wade 2011, Internet)

- (32) a. Wejdź na pokład **póki** możesz.
 come.IMPER on board as-long-as can.2SG
 ‘Get on board while you can.’
 b. Żyli długo i szczęśliwie, **póki** nie
 lived-PST.3PL long and happy until NEG
 umarli.
 died.PST.PF.3PL
 ‘They lived happily ever after until they died.’

(Polish: Internet)

- (33) a. **Pokiaľ** žijeme, existuje nádej.
 as-long-as live.1PL exists hope
 ‘As long as we live, there is hope.’
 b. Čítal, **pokiaľ** nezaspal.
 read.PST.IMPF POKIAĽ NEG-fall.asleep.PST.PERF
 He was reading until he fell asleep. (Slovak: Internet)

While the choice of preposition does affect the ambiguity in the clausal domain - *poka*, *pokí* and *pokiaľ* all allow for both the durative and the delimitative interpretation - it does provide us with a potential explanation for it. The key, I will argue, lies in the fact that *po*, unlike Czech *do* (cf. (2)), can function as an ambiguous marker in the nominal domain of all the languages mentioned above. That is, depending on whether the preposition assigns locative or accusative case, it can mean either *along* or *up to*, as is shown by the examples in (34)-(36).

- (34) a. Po dorog-e
 PO road-LOC
 ‘Along the road.’
 b. Po **pojas**
 PO waist.ACC
 ‘Up to the waist’ (Russian: Biskup 2019)
- (35) a. Po chodnik-u
 PO pavement-LOC
 ‘Along the pavement’
 b. Po kolan-a
 PO knees-ACC
 ‘Up to the knees’ (Polish: Biskup 2019)
- (36) a. Po ulic-i
 PO road-LOC
 ‘Along the street.’
 b. Po ulic-u
 PO street-ACC
 ‘Until the street.’ (Slovak: Mačutek, p.c.)

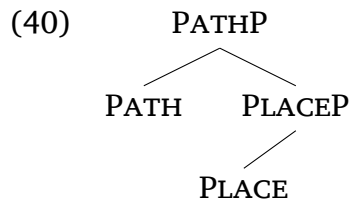
Even though the alternation exhibited by *po* in (34)-(36) is relatively restricted in its distribution, it fits with a broader well-documented pattern displayed by other case-sensitive adpositions in Slavic (and other) languages (cf. den Dikken 2003, Zwarts 2006, Biskup 2019 amongst others). Take, for instance, *w/v* ‘in’ and *na* ‘on’ in Russian, (37) and Polish, (38).

- (37) a. v / na stol-e
in / on table-LOC
b. v / na stol
in / on table.ACC
- (38) a. w / na stol-e
in / on table-LOC
b. w / na stół
in / on table.ACC

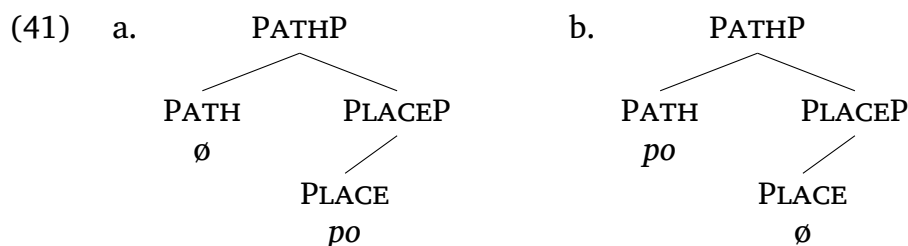
As with *po*, the interpretation of the prepositions in (37)-(38) depends on the case that is assigned to the accompanying noun: the locative case prompts a static, spatial reading, and the accusative case a directional one. The difference between these readings is less apparent when the prepositional phases (PPs) are compared in isolation, but it clearly surfaces when they are combined with stative verbs, such as *leżat’/leżeć* ‘lie’. While the PPs containing a noun in the locative case are compatible with these verbs, the ones with a noun in the accusative are not, as shown in (39).

- (39) a. leżat’/leżeć’ w / na stol-e
lie in / on table-LOC
b. *leżat’/*leżeć’ w / na stol/stół
lie in / on table.ACC

While the precise implementation of the case assignment in these constructions remains up to debate (see for instance den Dikken 2003, Fábregas 2007, Arsenijević & Gehrke 2009, Caha 2010, Haselbach & Pitteroff 2015 amongst others for various analyses of the prevalent dative vs. accusative case alternation), there is a general consensus regarding the formal representation of spatial PPs, as the ones in (34)-(39). With respect to the (morpho)syntactic structure, the by-now commonplace assumption is that it minimally consists of a stative Place projection and can be extended by a dynamic Path projection. This structure also reflects the semantics of locatives and directionals, with the former denoting a region or a set of points in space, and the latter an ordered set of points that define a trajectory from one location to another (cf. Jackendoff 1983, Zwarts 1997 et seq., Koopman 2000, Huybregts & van Riemsdijk 2002, Svenonius 2010, den Dikken 2010 amongst many others). The structure is depicted in (40).



Po (and by extension the other prepositions in (36)-(37)), as an ambiguous marker capable of expressing both locative and directional readings, would then be traditionally viewed as the overt exponent of either Place (in line with Koopman 2000) or Path (in line with den Dikken 2010).⁵ The other head would generally be realised by a silent or null lexical item in each account. Both possibilities are shown in (41a-b).⁶



However, both structures in (41) have been criticised for making inaccurate predictions. With regards to the structure in (41a), for instance, den Dikken (2003) notes that it wrongly predicts that several directional prepositions, such as Dutch *naar* ‘toward’, occur under a locative reading as well, because it assumes that all prepositions are base-generated in Place. With regards to the structure in (41b), on the other hand, Pantcheva (2011) remarks that it does the inverse, that it needlessly relies on a silent locative which otherwise never surfaces.

⁵Although the claim that case-sensitive adpositions are themselves ambiguous, is widely accepted (cf. Koopman 2000, Huybregts & van Riemsdijk 2002, den Dikken 2003, Pantcheva 2011 amongst others), it is not not entirely uncontested. Gehrke (2007), for instance, points out that (Germanic) locative prepositions can only be understood directionally under certain conditions (e.g. case-marking, specific verb classes, additional prepositions (*in-to*) etc.), suggesting that the ambiguity does not seem to be lexically encoded. Despite these valid remarks, I still opt to follow Pantcheva (2011) in this chapter, as her account does not stand in the way colexicalisation with other elements in the clause. I will, however, not further develop such an idea here, but leave it to future work which focuses more on the clause in its entirety.

⁶It is important to acknowledge that the structures in (41) are simplified depictions of the accounts by Koopman (2000) and den Dikken (2010). Especially with regards to the latter, I should clarify that he argues for an additional ‘Path’ projection, which he refers to as P_{DIR} . In his view, then, directional prepositions are inserted in the extended Aspect projection of this P_{DIR} . The main purpose of showing the structures in (41), however, is not to lay out Koopman and den Dikken’s analyses in full detail, but rather to convey that neither of them would consider *po* to be an exponent of both Path and Place, at least not at the same time.

Given these, and more general concerns about null morphemes, Pantcheva (2011) (also discussed in Rizzello, in this volume) rather explores the possibility that the structure is strictly compositional, i.e., that Place is contained within Path, and that prepositions like *po* are specified for both features, allowing them to lexicalise both syntactic layers via cumulative exponence - more commonly referred to as ‘phrasal lexicalisation’ in Nanosyntax (cf. Caha et al. in press, Starke 2009).

She finds empirical evidence for both of those claims in the different cross-linguistic patterns of syncretism between Path and Place exponents. Some languages like Tobati, (42), for instance, express the two meanings by means of individual, but morphologically contained lexical items, confirming the idea that Path builds on Place - especially because she does not encounter examples to the contrary. Other languages, however, like Bagvalal, (43), achieve the same thing by using just one, syncretic form, suggesting that some languages can lexicalise the same structure without requiring a 1:1 mapping between the number of morphemes and the number of features.

- (42) a. Ntric tad-i nanac.
 3PL sea.LOC swim.PL
 ‘They are swimming in the sea.’
 b. Nyiu tad-i-d rar.
 coconut sea-LOC-ALL fell
 ‘The coconut fell into the sea.’
- (43) beq’i
 barn-LOC/GOAL
 ‘in the barn’ or ‘into the barn’

From a technical perspective, the proposal for *po* then proceeds as follows. As her analysis of directional expressions is couched in Nanosyntax, Pantcheva adopts the view that syntactic structures are merged prior to lexicalisation, and that, as a consequence of late insertion, lexical items are conceived of as elements which connect those syntactic structures with phonological representations (and conceptual information, if there is any). As we have seen, *po* is a directional and locative expression, so it can be assumed that the lexical item for *po* includes the phonological representation /po/ and a syntactic structure which corresponds to a Path and a Place. The lexical item is shown in (44). Each feature represents a full phrase of the form XP.

- (44) /po/ ⇔ [PATH PLACE]

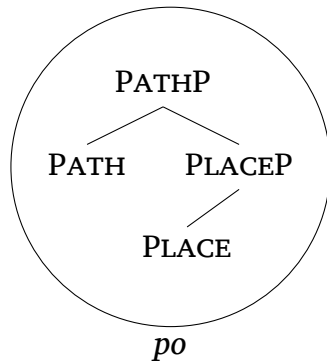
In this framework, lexicalisation is then understood as a matching process that is subject to an identity condition, better known as the Condition on Matching (Starke 2009), formulated here in (45).

(45) *Condition on Matching*

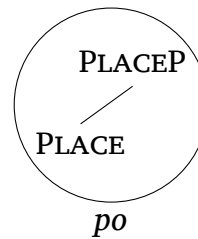
A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node.

This condition makes it possible for a lexical item to lexicalise a syntactic structure if it is fully identical to it, or if there is a partial identity with a subconstituent that is contained within it (often referred to as the Superset Effect). To visualise this, consider the trees in (46). In (46a), the lexical item of *po*, which consists of Path and Place, can be inserted at the height of Path and lexicalise both features, because it is a perfect match to the syntactic structure. In (46b), however, the syntactic structure just consists of Place. As indicated by the circle, though, *po* is still a match, because Place is a subconstituent of Path, and partial matches are permitted.

(46) a. DIRECTIONAL



b. LOCATIVE

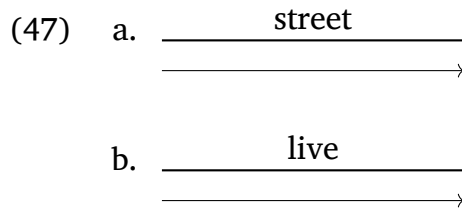


As we can see, Pantcheva's approach makes it possible to account for the observed distribution of *po* in both directional and locative contexts in a rather efficient way. The syncretism is simply captured by the assumption that there is an underlying structure which consists of two embedded syntactic layers, and which can be lexicalised separately or together due to the identity condition on matching and phrasal lexicalisation. In the following sections, I will continue to adopt this Nanosyntactic model of lexicalisation.

3.2 *Po* as a temporal marker

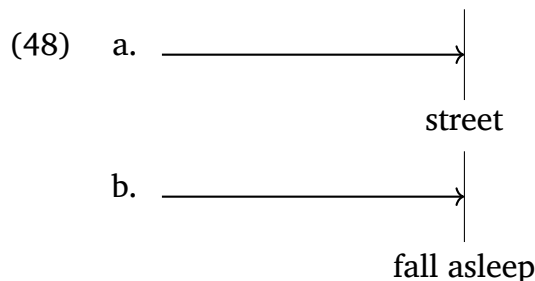
At this point, the question is how the use of *po* in the spatial domain is related to its use in the clausal and/or temporal domain. Meaning-wise, the contribution of the preposition seems to be comparable in

both domains. It either enables the physical placement of a Figure onto a Ground, or the figurative ordering of the matrix ‘Figure’ event with respect to the embedded ‘Ground’ event (see Talmy 2000, Svenonius 2010 and Caha in this volume, amongst others, for a detailed discussion on the concepts of Figure and Ground). Consider, for instance, the depictions of the Slovak locative *(Oni) kráčali po ulici* ‘(They) walked along the street’ in (47a), and of the durative *Pokiaľ žijeme, existuje nádej* ‘s long as we live, there is hope’ in (47b). The arrows represent the function of the preposition/conjunction *po*.



In (47a), *po* takes the Figure *they* and locates it onto the Ground *street*, and this in such a way that the Figure occupies all the points that are part of it (in contrast to other locative prepositions, such as *na* or *v* which only select a single point on the surface of that location). In a similar vein, *po* takes the matrix event *lit: there exists hope* in (47b) and ‘locates’ it onto the Ground event *we live*. That is, it coordinates and arranges them in a specific way with respect to each other. In this case, *po* treats the Ground event as an object consisting of multiple points as well - just points in time rather than in space - and the matrix event spans across the ‘length’ of its time interval, thus also occupying every time point that is contained within it (cf. the subinterval property in Dočekal 2012).

An analogous relationship appears for the directional and delimitative interpretations. Consider the depiction of the Slovak accusative *(Oni) čakali po ulicu Janka Kráľa* ‘People waited up to the Janka Kráľa street’ in (48a), and of the delimitative *Čítal pokiaľ nezaspal* ‘He was reading until he fell asleep’ in (48b).



In (48a) *po* again takes the Figure *they* and locates it with respect to the Ground *street*. However, this time, the Ground corresponds to a single

point which acts as a boundary for the path that *po* establishes - the path goes up to the street, but does not extend beyond it. The Figure thus also does not occupy any points of the location itself. Likewise, in (48b), *po* selects the matrix event *he was reading* and ‘locates’ it with respect to the Ground event *he fell asleep*. Just as for the spatial location, *po* treats the Ground event as consisting of a single ‘point’ in time which acts as an impenetrable border beyond which the matrix event cannot reach.

This correlation also gains support by the morphological facts. As became clear, Russian, Polish and Slovak present us with a syncretic pattern in both domains, as is summarised in the table in (49). In the spatial sense, we get an alternation between a length-in-space and a delimitative reading, whereas in the temporal one, we rather get an alternation between a durative and delimitative reading. Importantly, every single one of these readings is realised by the marker *po*.

(49)	SPATIAL TEMPORAL	
	NP	CLAUSE
DUR/DIST	<i>po</i>	<i>po</i>
DELIM	<i>po</i>	<i>po</i>

As was discussed in the previous subsection, the syncretism in the spatial domain is understood to arise because a complex underlying structure consisting of Path and Place feature is realised by a single exponent. If we take this seriously, it seems reasonable to treat the syncretism in the temporal domain in the same way. That is, *po*_{TEMPORAL} is also a lexical item which is able to lexicalise two syntactic features. If we then also take into account the functional similarities of *po* that were pointed out above, those layers likely correspond to Place and Path as well, with the former yielding a durative reading and the latter a delimitative one.

In arguing this, I must assume that there is a transfer from the spatial to the temporal domain, such that the properties found in the former domain are carried over to the latter one. This is not an unusual take. As discussed in Haspelmath (1997), the conceptual transfer of spatial notions to temporal ones is so incredibly widespread across languages, that it is generally assumed they are linked on a cognitive level, with temporal expressions clearly deriving from spatial ones (see also Kortmann (1997) on the unidirectionality of the transfer). The transfer itself is then conceived of as a kind of metaphorical shift. That is, the denotation of the spatial expressions are extended conceptually to incorporate the temporal ones.

Summarising, I discussed the *po*-pattern in Russian, Polish and Slovak, focusing first on its use in the nominal domain. As a spatial preposition, it was shown that *po* exhibits the well-known ambiguity between a

locative and a goal-oriented directional, two meanings commonly formalised as Place and Path. Following Pantcheva, I adopted the view that those two projections are hierarchically ordered and contained within each other, with Path containing Place, and that they can be lexicalised by means of a single lexical item through phrasal lexicalisation in Nanosyntax. Turning to the clausal domain, I argued that *po-* retains a similar function, expressing a temporal locative in the case of the durative *as long as*, and a temporal directional in the case of the delimitative *until*. Taking this correlation seriously, I proposed that a transfer took place from the spatial to the temporal domain, and that the syncretism between *as long as* and *until* falls out from the same mechanism - *po* is an exponent of Path and Place, now interpreted in a temporal sense rather than a spatial one.

In the next and final section, I will return to Czech *do-(kud)* and show that the proposal is also consistent with it.

4 Czech *dokud*

As things stand, the current proposal seems to hinge on the fact that the Russian, Polish and Slovak clausal marker *po* (*poka*, *pokí*, *pokial'*) has an identical counterpart in the nominal domain which is syncretic between a locative and a directional reading. This supports the idea, after all, that its lexical item contains a Place and Path feature that can be carried over to the temporal domain. However, this kind of reasoning raises questions with respect to Czech *do*, because, as we saw in the Introduction, this preposition never independently occurs under a durative, or spatial locative reading in the nominal domain. The nouns it co-occurs with do not exhibit any case alternations either which could potentially trigger such readings - they only appear in the genitive case. This is shown in (50), repeated from (2).

- (50) a. Léto vydrží **do** středy, potom se
 summer last DO Wednesday.GEN after REFL
 zatáhne a ochladí.
 cloud.over and cool.down
 'Summer will last until/*during Wednesday, afterwards it
 will become cloudy and cold.' [TEMPORAL]
- b. Musí ukrajinské děti chodit v Česku **do**
 have.to.3PL Ukrainian children go.INF in Czechia DO
 školy.
 school-GEN
 'Do Ukrainian kids have to go to/*along school in the Czech
 Republic?' [SPATIAL]

To express temporal duratives and spatial locatives which span a certain surface area, Czech rather employs the dedicated prepositions *během* ‘during’ and *po* ‘all over, through’, as illustrated by the examples in (51).

- (51) a. **Po** ulici.
all.over street.LOC
‘Along the street.’
b. **Během** pobytu.
during stay.GEN
‘During the stay.’

The full pattern in Czech is summarised in the table in (52). As can be seen, *do* is only syncretic in the clausal domain. In the nominal domain, it alternates with the other prepositions.

		TEMPORAL		
(52)		SPATIAL	NP	CLAUSE
	DUR DELIM	po	během do	do

Yet, the fact that there is no syncretism in the nominal domain does not necessarily pose a (technical) challenge for the analysis. Let me explain what I mean by that.

Starting again with the spatial domain, we see that *po* can only function as a locative marker (but see *infra* for a comment on this). This indicates that its lexical item likely only consists of a Place feature, as in (53a). In contrast, *do* is only used as a directional preposition. However, under the assumption that Path and Place are contained within each other, its lexical item cannot merely correspond to a Path feature - it must comprise both features, as shown in (53b). Recall that each feature below corresponds to a full phrase.

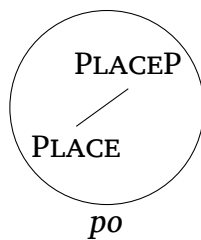
- (53) a. *po* ⇔ [PLACE]
b. *do* ⇔ [PATH PLACE]

In this set-up, however, both lexical items will be candidates to lexicalise Place, because, as the *Condition on Matching* in (45) stated, lexical items can lexicalise any syntactic (sub)constituent that is identical to (a part of) their own structure. To mediate this kind of competition between lexical items, the Elsewhere Condition applies in Nanosyntax. The formulation of this condition is given below in (54).

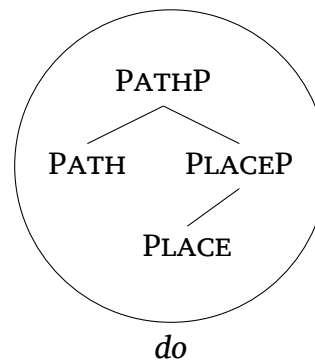
- (54) In case two rules, R1 and R2, can apply in an environment E, R1 takes precedence over R2 if it applies in a proper subset of environments compared to R2.

Put informally, the condition tells us that in a situation where two lexical items can be inserted, the one with the least amount of additional features will always win over the other. With this in mind, *po*, being the most suitable fit, will always realise Place, effectively preventing *do* from surfacing in locative environments. This is shown in (55a): *po* takes precedence over *do*, because it only contains Place. However, when Path is added to the structure, *do* can override *po* due to the containment of Place within its structure - that is, its lexical item is perfectly identical to that of a [Path[Place]] structure. This is visualised in (55b). The circles again indicate successful lexicalisation.

- (55) *po* $\Leftrightarrow$ [PLACE]
do $\Leftrightarrow$ [PATH PLACE]



- (56) *po* $\Leftrightarrow$ [PLACE]
do $\Leftrightarrow$ [PATH PLACE]



If we then assume that a spatio-temporal transfer took place, just like it did in Russian, Polish, and Slovak, the syncretism in the temporal domain falls out neatly again: *do* picks up the delimitative reading because it can lexicalise both Path and Place, and unlike in the spatial domain, there are no competing lexical items available for Place, so it can also pick up the durative reading.⁷ This is illustrated by means of the arrows in the table in (57). First, there is the conceptual transfer of *do* to the temporal domain. Second, the syncretism produces the ambiguity.

⁷This is also in line with findings by Langacker (1991). According to him, directional expressions can be used in durative ways, because they are able to express so-called 'abstract motions'. That is, in several contexts, they can refer to a location which spans along a line without invoking any physical motion (e.g. *There are wheat fields from the lake to the forest.*). As such, they are expected to be well-suited to express situations that extend over larger time intervals as well.

		SPATIAL	TEMPORAL
		NP	CLAUSE
(57)	DUR	po	do ↑
	DELIM	do →	do

Summing up, what the current proposal aimed to show is that the ambiguity between the durative and delimitative readings can also be derived without having to resort to complex, technical devices; the morphosyntactic decomposition of the subordinator in Czech (and by extension in Russian, Polish and Slovak) gets us there already without the interference of other clause-internal elements or syntactico-semantic operations.

That being said, there are still a few minor issues left with the analysis. In the remainder of this section I will briefly discuss them and outline possible directions to resolve them.

The first issue concerns the precise distribution of *po* in Czech. In specific cases, this preposition can also yield a directional reading. Consider the example in (58), where *po* with an accusative noun expresses *up to*.

- (58) Tankování nádrže až **po okraj** může způsobit
 fueling tank all-the-way PO rim.ACC may cause.INF
 poškození.
 damage
 ‘Filling your tank to the brim may cause damage.’ (Internet)

In this regard, one might wonder about two things: (i) why the conceptual transfer of *po* did not take place in Czech and (ii) how *po* and *do* are distinguished, considering they will both be Path markers.

Starting with the first question, the choice of which lexical items extend across different dimensions is, to a large extent, arbitrary. Think, for example, of adjectives, such as English *cool* and *warm* - while both receive an interpretation in the dimension of temperature, only *cool* crosses over to an evaluative dimension as well. Conversely, in French, for instance, the situation is reversed - *chaud* ‘warm’ may receive various evaluative readings, but not *froid* ‘cold’. It is thus expected that such cross-linguistic variations exist. What is stable in all cases is the fact that the transfer, if it takes place, is unidirectional.

With regards to the second question, it is likely that *po* will require an additional feature in its lexical structure. Not one related to Path, however, but probably one which is situated *in between* Place and Path. This to clearly set it apart as a locative preposition which refers to multiple points on a surface, instead of just a single point like *na* ‘on’. I leave it open for now what the exact nature is of this feature, but merging

it would essentially prevent *do* from realising the structure, because it would break up the constituency, and vice versa, not merging it, would allow *only do* to surface for the same reason (at least when one allows for backtracking, cf. Caha et al. in press).

A second issue pertains to the alternation between *během* ('during') and *do* in the nominal domain. While the analysis in its current state can handle this, it cannot constrain the distribution of *během* to *just* this domain. In order for it to be overridden by *do*, it must be considered as a locative Place marker. As such, its lexical item is smaller in size than that of *do*, and it becomes subject to the Elsewhere Condition. Given that this condition also holds in the clausal domain, it would be expected to appear there as well, contrary to fact. It can only be a part of the complex subordinating conjunction *během toho co* ('during it what/that'), as shown in (59).

- (59) Během toho co pracoval, byl okraden.
 during this.GEN that worked-he was-he robbed
 'While he was working, he was robbed.' (Internet)

Just like for *po*, I presume that a more fine-grained decomposition of the spatial PP structure will be required, as well as a better understanding of the importance of the instrumental marking *-em* (recall that Slovak in (8) also marked the temporal clauses with the instrumental) in order to accurately capture the distribution of this preposition. Given that the instrumental is often used to express bounded and unbounded motions *through* Ground locations (e.g. *po les-em* ('through the forest') vs. *po dveř-ma* ('through the door'), it is possible that the distinction is not between Place and Path, but rather between Paths and Routes (cf. Pantcheva 2011).

5 Conclusion

In this chapter, I discussed a curious case of syncretism exhibited by the subordinating conjunctions of temporal *as long as* and *until*-clauses across Slavic (and other) languages. I mainly focused on the one occurring in Czech. As was shown, the syncretism in this language could not be straightforwardly explained due to the fact that the durative reading does not follow from the meaning of the core components of the subordinator - that is, it does not seem to stem from the preposition *do*, which exclusively means *up to*, and much less from the relativising wh-element *kud*.

Previous accounts of (similar) facts have not the syncretism as such, but rather as a case of semantic identity. That is, they assumed a single, default reading that comes with a certain set of properties, and from this reading, they derived the other one, either through the interaction

with negation and aspect or through syntactic movement. While these accounts provided an explanation for how the durative and delimitative readings may arise, they were also shown to make wrong predictions with regards to Czech, and to run into technical issues to obtain the other reading in this language.

As an alternative, I then proposed an account that builds on the properties of the morphological components themselves. To lay the groundwork, I turned to Russian, Polish and Slovak. In these three languages, the clausal marker *po* also appears in the nominal domain, where it shows a different type of alternation - not between durative and delimitative readings, but rather between locative and a directional ones. Unlike the temporal alternation, this one is very well-known in the literature and typically captured (morpho)syntactically by means of a Path and Place projection. Following Pantcheva (2011), I adopted a Nanosyntactic approach to lexicalise this spatial PP structure. By specifying the lexical item of *po* for both features, and leaning on the concept of phrasal lexicalisation, I showed how the syncretism in the nominal domain could be explained as the realisation of two distinct syntactic layers by a single exponent.

After establishing this, I argued that the same structure is essentially transferred to the temporal domain. I gave two reasons for this. First, I discussed how the base function of *po* is to locate a Figure onto a given Ground, and who this function can be extended to include Figure and Ground events. Second, I remarked that this shared function is also reflected by the morphological identity in both domains. Assuming a spatio-temporal transfer, the puzzling syncretism in the temporal domain could easily be explained as well. Just like in the spatial domain, we are dealing with the realisation of a Path and Place layer by means of the same exponent.

Finally, I discussed how this proposal applies to Czech *do*. Despite the fact that this marker is not syncretic for Path and Place in the nominal domain, it must still be an exponent of both features owing to the containment relationship of these projections. This does not cause any misalignments with the data, however, because the Elsewhere Condition ensures that *do* never surfaces as a locative; other locative-only prepositions, such as *po*, are available, so they will take precedence in the lexicalisation process. *Do* will eventually surface as a directional, because it forms a perfect match with the full spatial PP structure. Since its lexical item is specified for both features as well, the same spatio-temporal transfer as in Russian, Polish and Slovak, again guarantees the explanation of the syncretism.

As usual, several questions are left unanswered. The most pressing one, of course, concerns the presence of negation and, in the case of Czech, (im)perfective aspect in these temporal clauses. That is, if the distinction between a durative and a delimitative interpretation is reduced to a Path structure at the level of subordination, it becomes

unclear why the other clause-internal elements are required in this context, and another function ought to be attributed to them. Although I cannot offer any conclusive answers in this chapter, I tentatively want to propose that the combination of negation and verbal aspect might be expressing a form of modality - be it epistemic or evidential. This proposal is partly motivated by arguments made in earlier work on so-called ‘expletive’ negation, such as Makri (2013) and Tsiakmakis & Espinal (2022) amongst others, and partly by the existence of expletive negation markers in languages, like Chinese *mei*, which are exclusively associated with perfective aspect - a grammatical category that has itself been linked to evidentiality. Testing this hypothesis for Czech, in particular, will be challenging, given the absence of minimal pairs, but I hope to explore this avenue from a cross-linguistic perspective in future work and ultimately provide a uniform analysis of *as long as* and *until*-clauses as a whole.

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